



Universität
Zürich^{UZH}



Blockchain & DLT
Research Group



Network Science HS24

Assignment 7

Blockchain & Distributed Ledger Technologies Group

FOR STUDIES PURPOSES ONLY

UZH Blockchain Center
Faculty of Business, Economics and Informatics
University of Zurich
Zurich, November 4, 2024

1 Robustness

1. (1 point) Generate synthetic networks (you can use the software/package you prefer, as always we suggest you `networkx`):
 - Random networks (Erdos–Rényi) (with $n = 1000$ and $\langle k \rangle = \{1, 4\}$)
 - Scale-free networks (Barabási-Albert) (with $n=1000$ and $m = \{2, 4\}$)
2. (2 item) Plot the degree distributions of the synthetic networks and of the networks provided in the datasets.
3. (2 points) Plot the size of the largest connected component as a function of removed edges and nodes in two scenarios for the synthetic graphs and the real networks in the datasets:
 - (a) Random attacks (failures)
 - (b) Targeted attacks (degree centrality and betweenness centrality)
4. (1 point) Comment the plots, using what we discussed in class.

Datasets provided

- `graph_eu_airlines.gml`: European airline network. Nodes represent airlines and Edges represent air-line routes among European airports Cardillo et al. [2013]
- `graph_power.gml`: The file contains an undirected unweighted representation of the topology of the Western States Power Grid of the United States Watts and Strogatz [1998]

References

- Cardillo, A., Gómez-Gardenes, J., Zanin, M., Romance, M., Papo, D., del Pozo, F. and Boccaletti, S. [2013]. Emergence of network features from multiplexity, *Scientific Reports* **3**: Article number: 1344.
- Watts, D. J. and Strogatz, S. H. [1998]. Collective dynamics of ‘small-world’ networks, *nature* **393**(6684): 440–442.